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Faculty of Engineering — Mechanical Engineering

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Mass Unbalance in High-Speed Slip Ring Systems

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Contents

1. The Company and the Department

1.1 Foundation and Historical Background

1.2 Defence System Technologies Sector Presidency

1.3 Slip Ring Design Engineering Team

1.4 Scope and Objectives of the Internship

2. Slip Rings

2.1 The Rotary Power and Data Transfer Problem

2.2 Non-Contacting Technologies

2.3 Slip Ring Configurations

2.4 Contact Technology and Materials

2.5 Application Domains and Their Constraints

3. Project 1 — Balancing of a Slip Ring Rotor

3.1 Fundamental Concepts

3.2 Consequences of Unbalance

3.3 Determination of Tolerances

3.4 Translation into Manufacturing Tolerances

3.5 Rigid and Flexible Rotor Behaviour

3.6 The Plane-Count Decision

3.7 Measurement and Correction Methods

3.8 Component Description and Requirements

3.9 Balance Quality Grade Selection

3.10 Verification of the Rigid Rotor Assumption

3.11 Plane-Count Decision and the Inertia Criterion

3.12 Manufacturing and Assembly Error Budget

3.13 Balancing Machine Selection

3.14 Measurement Uncertainty and Acceptance

3.15 Correction Method Selection

3.16 Findings and Open Items

4. Project 2 — Design and Manufacture of an Ergonomic Tool Holder

5. References

1. The Company and the Department

1.1 Foundation and Historical Background

ASELSAN (Askeri Elektronik Sanayi, Military Electronics Industry) was established in Ankara in 1975 by the Turkish Armed Forces Foundation (TSKGV) [1]. The company was founded in direct response to the arms embargoes imposed on Turkey following the 1974 Cyprus Peace Operation, with the objective of meeting the communication requirements of the Turkish Armed Forces through national resources [1], [2]. Its formation was led by the Armed Forces Strengthening Foundations, which had been created through public donations during the 1970s [3].

Although the company initially concentrated on military radios and communication equipment, its scope has broadened substantially over the past five decades. ASELSAN today designs, develops and manufactures systems for land, air, naval, space and unmanned platforms, covering a range that extends from material-level design up to complete system integration [1]. Its principal areas of expertise include radar and electronic warfare systems, electro-optical systems, guidance technologies, avionics, and command and control systems [2].

The company is organised into sector presidencies according to the differing investment and production structures required by its project portfolio: Communication and Information Technologies (HBT), Microelectronics, Guidance and Electro-Optics (MGEO), Radar and Electronic Warfare Systems (REHİS), Defence System Technologies (SST), and Transportation, Security, Energy and Automation Systems (UGES) [3], [4]. Engineering and production activities are carried out at the Macunköy, Akyurt, Gölbaşı and Teknokent campuses in Ankara [4].

Beyond the defence sector, ASELSAN applies the technologies it develops to civil fields including transportation, energy, security and healthcare [1]. Supported by a research-intensive structure and an export-oriented growth strategy, the company is today among the largest technology organisations in Turkey [2].

1.2 Defence System Technologies Sector Presidency

The internship was carried out within the Defence System Technologies Sector Presidency (SST), one of the five sector presidencies through which ASELSAN organises its activities [3]. SST operates its own dedicated research and development centre in Gölbaşı, Ankara [5].

The scope of SST covers weapon and defence systems for land, air and naval platforms. Its product portfolio includes remote-controlled machine gun and grenade launcher stations, fire control systems for tanks, howitzers, mortars, multi-barrel rocket launchers and naval guns, smart munitions, and active protection systems designed to detect and neutralise incoming threats to land vehicles before impact [6]. On the air defence side, the presidency develops systems aligned with a layered air defence concept, integrating weapon and missile systems with high-performance sensors, command and control, communication, and fire control subsystems [7].

1.3 Slip Ring Design Engineering Team

The internship was carried out within the research and development design team working on slip ring systems. The scope of the team's work spans the full development cycle of this component: conceptual

design, prototype manufacture and verification testing. Activities include defining the electrical channel architecture, selecting brush and track materials, designing the mechanical housing, satisfying sealing and environmental durability requirements, specifying manufacturing tolerances, and defining post-assembly quality verification steps.

Because a slip ring carries the entire power and data connection of the rotating load it serves, it constitutes a single point of failure with no redundancy within the system. This characteristic forms the fundamental justification for the rigour applied throughout the design and verification processes.

1.4 Scope and Objectives of the Internship

The engineering problem assigned for the internship concerns mass unbalance in the rotor of a slip ring assembly. The component in question is a hollow aluminium rotor of 700 g intended to operate at a nominal speed of 1500 rev/min.

Unbalance arises from the separation between the geometric axis of rotation and the mass axis of inertia of a rotating body. This separation cannot be eliminated in manufacture, and the centrifugal force it produces grows with the square of rotational speed [8]. In a slip ring, the resulting force is transmitted through the bearings to the housing and modulates the contact pressure at the brush–track interface, so that the mechanical quality of the rotor bears directly on the electrical function of the component.

The assignment is structured around three questions: how unbalance is measured, what manufacturing precision should be specified, and by which method the measured unbalance is corrected. The study is bounded in three respects. It addresses a single operating point, so that the conclusions are specific to this rotor, this speed and this bearing arrangement. It is analytical, since no balancing measurement was performed on the physical component during the internship period. And two contributions to the error budget — the distribution of the internal wiring and the mass of the fixing and bonding elements — are engineering estimates rather than measured values.

In addition to the balancing study, a secondary design task was assigned: an ergonomic holder for hand tools such as screwdrivers and soldering equipment, intended to improve the arrangement of the assembly and soldering station in the laboratory. This task is described in Part 4.

2. Slip Rings

2.1 The Rotary Power and Data Transfer Problem

Any system containing an assembly that rotates relative to its supporting structure faces a common engineering problem: electrical power and data must cross the rotating interface without interruption. The subsystems mounted on the rotating assembly — sensors, actuators, control electronics, heating elements — require a supply, and the signals they generate must reach the stationary control unit.

Where angular travel is limited, the problem admits a trivial solution. A flexible cable harness of sufficient length can span the interface, accommodating the relative motion through its own elasticity. Its limitation is equally clear: the harness can absorb only a finite angular displacement before it is stressed beyond its permissible bend radius, and repeated flexing produces conductor fatigue and

insulation cracking, so that even within the permitted range the harness becomes a life-limited component.

The requirement that distinguishes the applications considered here is therefore not rotation as such, but unlimited continuous rotation. Once this requirement is imposed, the cable harness is eliminated as an option and a dedicated rotary interface device becomes necessary. Such devices fall into two families: non-contacting technologies, which transfer energy or information across an air gap, and contacting technologies, of which the slip ring is the principal example.

2.2 Non-Contacting Technologies

Four non-contacting technologies are in established use, and each is defined by the physical quantity it transfers across the gap. A capacitive rotary coupler forms a variable capacitance between facing electrode surfaces and transfers data at high rates, but the impedance of a practical air gap is far too high to supply a load, and the coupling varies directly with gap width so that rotor eccentricity appears as amplitude modulation. An inductive coupler, or rotary transformer, transfers substantial power by magnetic induction across a split ferromagnetic core without wear, but transfers alternating current only, so that a direct-current supply requires inversion on the stationary side and rectification on the rotating side. A fibre optic rotary joint maintains optical continuity between a rotating and a stationary fibre, offering the highest data capacity of any rotary interface and complete immunity to electromagnetic interference, but transferring no electrical power and demanding tight optical alignment. An RF rotary joint transfers radio-frequency energy through a coaxial or waveguide structure whose geometry is invariant under rotation, which makes it indispensable in radar installations, but each RF path requires its own structure and the device transfers neither direct current nor low-frequency data.

Technology	Unlimited rotation	Power transfer	Multi-channel data	Wear
Cable harness	No	Yes	Yes	Fatigue-limited
Capacitive coupler	Yes	No	Limited	None
Rotary transformer	Yes	AC only	Limited	None
Fibre optic rotary joint	Yes	No	Limited	None
RF rotary joint	Yes	No	One per path	None
Slip ring	Yes	Yes	Yes	Sliding contact

Capability of the available rotary interface technologies against the three requirements that define the problem.

No non-contacting technology satisfies all three requirements simultaneously. Each excels within a narrow domain and must be supplemented by a further device to deliver a complete interface. The slip ring is the only technology that combines unlimited rotation, direct- and alternating-current power transfer, and an arbitrary number of independent data channels within a single mechanical interface. This capability accounts for its continued use despite the disadvantage it carries, namely a sliding electrical contact subject to friction, wear and contact resistance variation. The performance of the complete device is governed by the stability of that contact, and every subsequent section of this report addresses one aspect of maintaining it.

2.3 Slip Ring Configurations

Slip rings are classified by the geometric arrangement of their conducting tracks and by the service environment for which they are designed. The configuration determines the available channel count, the envelope of the device, the achievable sliding velocity and, as shown in Section 2.3.4, the mass distribution of the rotor.

2.3.1 Centrifuge type

In the centrifuge configuration the conducting tracks are arranged as concentric rings on a rotor of comparatively large diameter, and the assembly is designed to operate under the elevated radial acceleration associated with centrifugal machinery and large rotating platforms. The configuration is used predominantly for power channels: the large track diameter provides the circumferential length required to accommodate wide brush contact areas, and wide contact areas are what permit high current density to be carried without excessive local heating. Correspondingly, this type operates at low rotational speed, since the sliding velocity at the contact is the product of track radius and angular velocity.

Two design consequences follow. Brush contact force must be specified with the radial acceleration field in mind, since the brush and its spring possess mass and are themselves subject to that field. And rotor balancing becomes a primary design constraint rather than a finishing operation, because the centrifugal force produced by a given unbalance scales with both the rotor mass and the square of angular velocity.

2.3.2 Pancake type

In the pancake configuration the tracks are arranged as concentric circles on a single flat disc, and the brushes contact the disc face rather than a cylindrical surface. This arrangement is selected when the axial envelope is severely constrained and radial space is available; adding a channel increases the diameter of the device rather than its length.

The configuration carries an inherent asymmetry that has no counterpart in cylindrical designs. Each track lies at a different radius, so the sliding velocity, the contact path length per revolution and consequently the wear rate differ from channel to channel. The wear allowance of the assembly is governed by the outermost track, and channel allocation must therefore take radius into account.



2.3.3 Vacuum type

The vacuum configuration is a slender, axially extended assembly intended for signal transmission in evacuated or near-evacuated environments. Its geometry is dictated by the channel count: tracks are stacked along the axis, separated by insulating spacers, so the device grows in length rather than in diameter.

Environmental constraint. Carbon brushes rely on a graphite film that forms on the track surface during operation, and the formation of that film depends on adsorbed water molecules; in the absence of ambient humidity the film cannot form and the wear rate increases by orders of magnitude [13], [14]. This behaviour is examined in Section 2.4.3. Vacuum designs therefore employ additive-containing brush materials, in which compounds such as molybdenum disulphide substitute for the lubricating role of adsorbed water, together with low-outgassing insulating materials selected to prevent volatile species from condensing on the contact surfaces.

Construction. The rotor is built around a tubular body carrying a stack of conducting tracks separated by insulating rings. Each track is provided with an integral termination lug projecting into the inner cavity of the rotor. The lug provides a defined location at which the internal conductor is joined to the track, so that the joint is made away from the sliding surface; it provides mechanical strain relief, so that any load transmitted along the conductor is reacted at the lug rather than at the track; and it establishes a repeatable routing geometry for the conductor, which is significant because the axial and angular position of the internal wiring contributes directly to the mass distribution of the rotor.

2.3.4 Effect of configuration on rotor geometry

The configurations produce rotors of markedly different proportions. The centrifuge type yields a rotor of large diameter and short axial length; the pancake type an extreme of the same tendency; the vacuum type a slender rotor whose length increases with channel count. Expressed as the ratio of rotor length to rotor diameter, a pancake or centrifuge rotor has a low ratio, whereas a vacuum-type rotor may have a ratio well above unity.

This ratio is not merely descriptive. In rotor balancing it is conventionally used as the criterion by which a rotor is classified as disc-like or shaft-like, and this classification determines whether unbalance may be corrected in a single correction plane or whether two correction planes are required [9]. The configuration decision taken at the concept stage therefore constrains the balancing procedure applied at the end of manufacture. Section 3.6 examines this criterion in detail and Section 3.11 demonstrates that the conventional threshold is not a fixed value but depends on the cross-sectional geometry of the rotor.

2.4 Contact Technology and Materials

2.4.1 Precious metal wire brushes

A precious metal wire brush consists of one or more fine wires, typically of gold, silver or palladium alloy, cantilevered against the track surface. The wire performs two functions simultaneously: it carries the current and, through its own bending stiffness, generates the contact force, so that no separate spring element is required. Two characteristics make this construction suitable for signal channels. The contact resistance of a noble metal pairing is low and, more importantly, stable, since neither surface forms an insulating oxide. And several wires are used per channel, which provides redundancy at the contact level: momentary loss of contact by one wire does not interrupt the circuit because the remaining wires maintain continuity [11], [19].

The limitation is current capacity. The cross-section of a fine wire and the small real contact area restrict the permissible current, so this construction is not used for power channels. Material selection within the noble metals is a compromise between conductivity, hardness and cost: pure gold and pure

palladium are soft and prone to material transfer under sliding [18], so hardened alloys are used in practice, and cladding or selective plating limits the quantity of precious metal required [12].

2.4.2 Carbon and metal–graphite brushes

A carbon brush is a monolithic block of graphite, or of graphite compounded with a metallic phase, pressed against the track by a separate spring. The graphite provides lubrication at the sliding interface while the metallic phase provides the conduction path [15], [19]. Metal–graphite brushes are the standard solution for power channels, since the larger contact area and the bulk conductivity of the compound permit current densities far beyond the capability of a wire brush.

The choice of metallic phase depends on the duty. Copper–graphite is the most widely used general-purpose composition, while silver–graphite is preferred where signal quality, high current density or high sliding velocity is required, since silver offers higher conductivity and its oxide remains conductive [16], [19]. Brush wear and track wear are strongly coupled and the material pairing must be selected as a pair; experimental studies have shown that metallurgical incompatibility between the track metal and the brush graphite minimises wear, which is why the combination of a copper or copper-alloy track with a silver- or copper-graphite brush has become established practice [19], [20].

2.4.3 Graphite film formation and environmental dependence

The tribological behaviour of a carbon brush is governed not by the bulk properties of the graphite but by a thin transfer film that develops on the track surface during the initial running period. This film acts simultaneously as a solid lubricant and as a conducting layer, and its presence is what permits a carbon brush to operate for extended periods at acceptable friction and wear rates.

Film formation depends on the presence of adsorbed water molecules, which intercalate between the graphite lamellae and reduce the shear resistance between them. The governing quantity is relative humidity rather than absolute humidity, since the surface coverage attained by an adsorbed species is determined by the ratio of the partial pressure of water vapour to its saturation pressure at the prevailing temperature. Below a threshold of a few percent relative humidity the film cannot be sustained and the brush enters a regime commonly termed dusting, in which material is removed as loose particles rather than transferred as a film, and the wear rate is higher by orders of magnitude. Two operating conditions produce it: vacuum, where water vapour is absent by definition, and high altitude, where the partial pressure of water vapour falls with the total ambient pressure.

The engineering response is the incorporation of additives — molybdenum disulphide, barium fluoride and similar compounds — which provide lamellar lubrication without requiring adsorbed water. A second consequence of dusting is electrical rather than mechanical: the detached particles are conductive, and their accumulation within the enclosed volume of the device reduces the insulation resistance between adjacent channels. Particle generation is therefore a matter of electrical safety and not only of service life.

2.4.4 Track materials and plating

The track is the rotating counterface of the sliding pair, and since no single metal satisfies the requirements of conductivity, hardness, corrosion resistance and compatibility with the brush material simultaneously, tracks are almost always built as a plated structure on a base material. Copper forms the base of most constructions: its conductivity is second only to silver among practical engineering metals and copper-alloy tracks running against metal–graphite brushes constitute the established

pairing for power channels [19], [20], although bare copper forms an insulating oxide in service and is therefore unsuitable as an exposed surface for low-level signals. Nickel is deposited as an intermediate barrier layer to prevent interdiffusion of copper into the gold surface and to provide a harder substrate beneath the comparatively soft surface plating. Silver is used both as a surface plating and as the metallic phase in composite brushes; its distinguishing advantage in sliding contact is that silver oxide remains electrically conductive [16], and its disadvantage is susceptibility to sulphide tarnishing in contaminated atmospheres. Gold is applied as the surface plating of signal tracks because it does not oxidise, so that contact resistance remains stable over long service intervals; this stability rather than conductivity is the reason for its selection, and since pure gold is soft, hard gold alloyed with cobalt or nickel is used where sliding is involved. A representative signal track is accordingly a copper base carrying a nickel barrier layer and a hard gold surface layer, contacted by a gold-alloy wire brush, while a representative power track is a copper or copper-alloy base, unplated or silver-plated, contacted by a silver–graphite or copper–graphite brush.

2.4.5 Channel allocation and redundancy

The channels of a slip ring assembly are not interchangeable; each is assigned to a specific function, and the contact technology, track plating and brush material are selected accordingly.

Power channels carry the supply to motors, heaters and electronic units on the rotating assembly, and are allocated to metal–graphite brushes running on copper-alloy tracks, positioned where the thermal path to the housing is most favourable.

Data channels carry the digital and analogue signals of the rotating subsystems. The protocols encountered include CAN bus, RS-422 and Ethernet. The differential protocols require their conductor pairs to be allocated to adjacent channels so that induced disturbances appear as common-mode, while Ethernet additionally requires controlled impedance and shielding, since the contact discontinuity represents an impedance mismatch in the transmission line.

Redundant channels are provided as a matter of design practice. Spare tracks are included in the assembly and left unconnected, so that a channel exhibiting degraded contact resistance in service can be reassigned without replacing the device. Since the slip ring constitutes a single point of failure for the entire rotating assembly, the marginal cost of a spare track is small relative to the consequence of its absence.

2.5 Application Domains and Their Constraints

Three domains are examined below. Each is presented in the same form — the application, the function performed, and the constraint that dominates the specification — so that the comparison in Section 2.5.4 rests on a consistent basis.

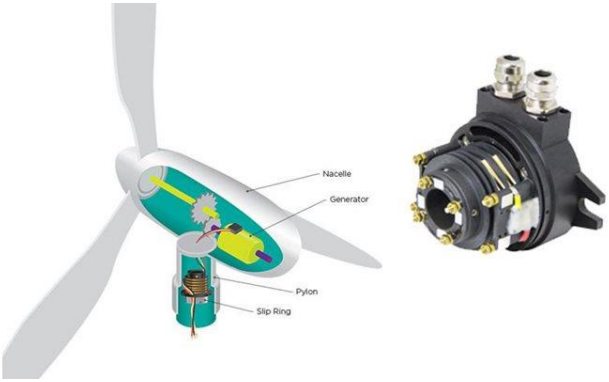
2.5.1 Wind turbine pitch systems

Application. In a utility-scale wind turbine the slip ring is located at the interface between the stationary nacelle and the continuously rotating hub, and is variously referred to as a pitch ring, pitch control slip ring or hub slip ring [21].

Function. Modern turbines adjust the pitch angle of each blade individually and continuously in response to wind speed and rotor speed. Each blade is driven by a dedicated actuator located inside the hub, and the slip ring provides both the power feed to these actuators and the digital

communication path — commonly CAN bus — by which the pitch controller commands them in real time [21], [24]. A momentary loss of electrical continuity is sufficient to raise a pitch control fault, which forces the turbine to feather its blades and shut down, so the reliability of this single component bears directly on generation availability [24]. Redundancy is accordingly designed into the assembly at system level [25].

Governing constraint. Operational life under harsh environmental exposure. The rotational speed is low, typically below twenty revolutions per minute, so neither sliding velocity nor rotor unbalance is demanding. What is demanding is the service interval: the assembly is mounted at hub height where access requires specialised equipment, and maintenance visits are scheduled at intervals of months [21].

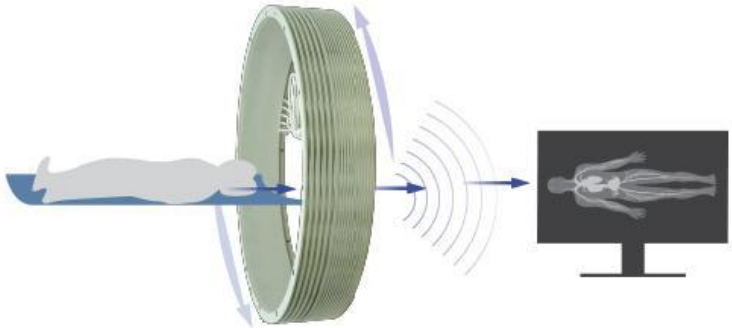


2.5.2 Computed tomography gantries

Application. In a third-generation computed tomography scanner the gantry rotates continuously about the patient, carrying the X-ray tube, the high-voltage tank and the detector array, while the gantry support remains stationary. The slip ring is positioned at the rotating base and forms the sole electrical connection between the two [23].

Function. The assembly carries power from the stationary inverter to the high-voltage tank and simultaneously returns the output of the detector array to the data acquisition system [23]. The introduction of the slip ring is what permitted continuous helical scanning to replace the earlier stop-and-reverse acquisition.

Governing constraint. Continuity of the data stream. Rotational speed is high, in the range of one hundred fifty to three hundred revolutions per minute, and the rotating mass is measured in hundreds of kilograms, so the centrifugal forces arising from unbalance are substantial. The dominant requirement, however, is that any momentary variation in contact resistance appears in the reconstructed image as an artefact, so the requirement is stated in terms of permissible contact resistance variation rather than wear life.



2.5.3 Stabilised turrets and remote controlled weapon stations

Application. In a remote controlled weapon station the slip ring is located on the azimuth axis, between the fixed base mounted to the platform and the rotating upper structure carrying the weapon and the sensor suite.

Function. The assembly carries the power supply and the data connections of every subsystem mounted on the rotating structure: the electro-optical sensor package, the firing and triggering circuits, and the supply and feedback paths of the stabilisation servos. The data traffic is mixed in character, combining low-rate control messages over CAN or RS-422 with high-bandwidth video over Ethernet. Without the slip ring the station could not complete a full revolution, and the alternatives are incompatible with continuous target tracking.

Governing constraint. Line-of-sight stability. The rotational speed is low to moderate and the duty cycle intermittent, so neither wear life nor thermal loading is limiting. What limits the design is the vibration the device transmits into the supporting structure: the rotating assembly carries an optical sensor whose angular stability determines system performance, and any disturbance introduced at the azimuth interface appears in the image as jitter. Unbalance in this application is therefore a mission performance concern rather than a maintenance concern. A second constraint arises from the platform, since land vehicles impose mechanical shock and broadband vibration, so that sealing must be maintained under these loads and all fasteners require positive locking.

2.5.4 Comparative assessment

Domain	Speed	Rotating mass	Maintenance interval	Governing constraint
Wind turbine pitch	Very low	Moderate	Months; difficult access	Wear life under exposure
Computed tomography	High	Large	Scheduled service	Data stream continuity
Stabilised turret	Low to moderate	Moderate	Platform cycle	Line-of-sight stability

Operating parameters and governing constraints across the three application domains examined.

The operating parameters span more than an order of magnitude in rotational speed, and since the centrifugal force produced by a given unbalance scales with the square of angular velocity, the same rotor represents a substantially different mechanical problem in each case. More significantly, the governing constraint differs not in degree but in kind: wear life, data continuity and transmitted vibration are not points on a common scale, and optimising for one does not improve the others.

This observation determines the approach taken in Part 3. Balance quality standards classify rotors by machine type and assign a grade to each class on the basis that machines of a given type share a characteristic speed range and a characteristic sensitivity to unbalance [9]. A slip ring satisfies neither premise: it is a component rather than a machine, it is installed across a speed range spanning more than an order of magnitude, and the constraint that determines its tolerance is inherited from the system in which it is installed. No balance quality grade is therefore defined for slip rings in the standard, and this absence is a consequence of the classification logic rather than an omission within it.

3. Project 1 — Balancing of a Slip Ring Rotor

3.1 Fundamental Concepts

3.1.1 Definition

Perfect balance is the condition in which the geometric axis of rotation of a component coincides spatially with its mass axis of inertia. This coincidence cannot be achieved in manufacture: no surface can be machined perfectly concentric, no material is perfectly homogeneous, and no assembly is perfectly symmetric. The two axes always separate, and unbalance is the measure of that separation [26].

The practical value of the definition lies in the fact that the manner in which the two axes separate determines the type of unbalance directly. If the axes are parallel but displaced, the unbalance is static; if they intersect but form an angle, it is couple unbalance; if both conditions apply, it is dynamic unbalance.

3.1.2 Eccentricity and unbalance

Eccentricity is the perpendicular distance from the centre of mass to the axis of rotation and has the dimension of length:

$$e = \sqrt{(x_s^{\,2} + y_s^{\,2})}$$

where x_s and y_s are the components of the radial displacement of the centre of mass with respect to the axis of rotation. Unbalance is the product of eccentricity and mass:

$$U = m\, e$$

Its dimension is mass times length, and it is expressed in practice in g·mm. The two quantities serve different purposes: eccentricity is independent of mass and is used when specifying manufacturing tolerances, while unbalance is used when calculating the correction mass.

3.1.3 Types of unbalance

Static unbalance. The mass axis of inertia is parallel to the axis of rotation but radially displaced from it. When the rotor is placed at rest on frictionless bearings, the heavy region rotates to the lowest position under gravity; the term derives from this behaviour. The magnitude and angle are obtained from the sum of the unbalance vectors measured in the two correction planes:

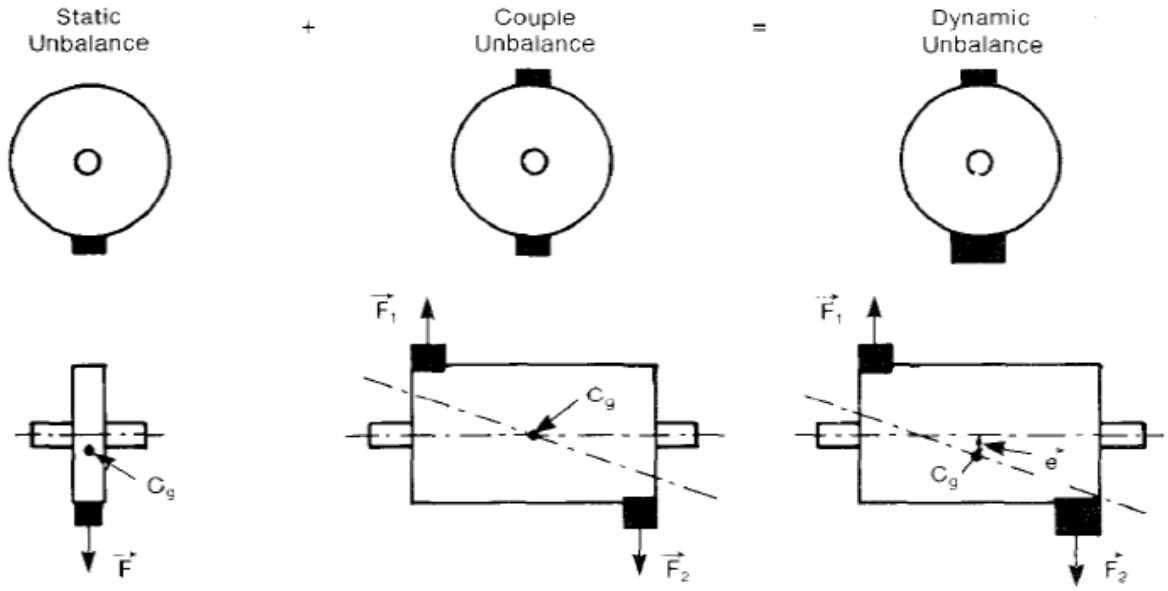
$$U_s = U_L + U_R\,, \quad \varphi_s = \arctan\,(y_s \,/\, x_s)$$

Couple unbalance. The centre of mass lies on the axis of rotation; the error is not a displacement of the mass but an asymmetric distribution of it along the axis. Under a gravity test the rotor appears balanced and the error becomes apparent only in rotation. Couple unbalance is expressed in terms of the products of inertia of the mass distribution:

$$U_c = (I_{xz}\,, I_{yz})\,, \quad |U_c| = \sqrt{(I_{xz}^{\,2} + I_{yz}^{\,2})}\,, \quad \varphi_c = \arctan\,(I_{yz} \,/\, I_{xz})$$

Here $I_{xz} = \int x\, z\, dm$ and $I_{yz} = \int y\, z\, dm$ are the mass products of inertia. A non-zero product of inertia indicates that the axis about which the body is rotated is not a principal axis, which is the precise equivalent of couple unbalance.

Dynamic unbalance. The superposition of the static and couple components. Almost all rotors produced in practice fall into this category.



3.1.4 Difference in units

Static unbalance is the first moment of the mass distribution, whereas couple unbalance is its second moment, so that their units differ and they cannot be compared directly. To permit comparison, the couple component is divided by the distance between the correction planes; the quantity U_c / b has the dimension g·mm and may be compared against the plane tolerance.

3.1.5 Force and moment relations

In the general case, where the angular acceleration is non-zero:

$$F_s = U_s \sqrt{(\alpha^2 + \omega^4)}, \quad M = U_c \sqrt{(\alpha^2 + \omega^4)}$$

The ω^4 term is the contribution of the centrifugal force acting in the radial direction, while the α^2 term is the contribution of the Euler force arising from the tangential acceleration of the displaced mass. Since the two forces are mutually perpendicular, the resultant is their root-sum-square. In steady-state operation $\alpha = 0$ and the expressions reduce to $F_s = U_s \omega^2$ and $M = U_c \omega^2$.

Static unbalance produces a net force and no net moment, so that both bearings are loaded in the same direction and in phase. Couple unbalance produces zero net force and a non-zero net moment, so that the bearings are loaded in opposite directions:

$$F_A = F_B = U_s \omega^2 / 2, \quad F_A = -F_B = U_c \omega^2 / L_b$$

The negative sign is the origin of the phase difference observed in measurement and forms the basis of the plane-count verification criterion described in Section 3.6.4.

3.2 Consequences of Unbalance

The rotating force produced by unbalance affects system performance through four mechanisms, whose relative significance varies with the application and must be established by calculation.

Bearing load and fatigue life. The unbalance force passes through the bearings, and in ball bearings the nominal life is proportional to the cube of the ratio of dynamic load rating to equivalent dynamic load, $L_{10} = (C / P)^{3}$ [27]. A twofold increase in load therefore reduces life to one eighth, which is the reason unbalance has traditionally been treated as a bearing life problem.

Vibration transmitted to the supporting structure. The force transmitted through the bearings to the housing consumes part of the vibration budget of the structure on which the assembly is mounted, and its distinguishing characteristic is that unbalance produces a purely first-order excitation, a single spectral line at the rotational frequency. A narrow-band continuous excitation coinciding with a structural natural frequency produces resonance, in which the response is amplified even when the source itself is weak.

Uniformity of wear. The rotating force modulates the contact pressure at the brush–track interface in the circumferential direction, and since the wear rate depends on contact pressure, one region of the track wears more rapidly than another. The modulation is synchronous rather than random, recurring at the same angular position on every revolution, so that the cumulative effect produces a measurable profile on the track surface over a period of months.

Contact resistance fluctuation. An inverse relation exists between contact resistance and contact force; in the plastic deformation regime the real area of contact is proportional to the force, so that R_c ∝ F^(−γ) and $\Delta R / R = - \gamma (\Delta F / F)$ [13]. Every fluctuation in force is therefore converted into a fluctuation in resistance, which appears as noise on signal channels. The relation is derived for static contacts and does not account for the surface film resistance that dominates in carbon brushes [14], and the exponent is reported experimentally between 0.2 and 1.

The attenuation chain. Unbalance does not reach the brush contact directly but through a chain: centrifugal force, bearing deflection, rotor orbit, runout at the contact point, and fluctuation in contact force. The bearings and the brush spring behave as parallel springs supporting the rotor at a common displacement, so that the load is shared in proportion to stiffness, $x = F / (2k_b + k_f)$. Since the radial stiffness of a preloaded bearing and the spring rate of a brush wire differ by orders of magnitude, the displacement of the rotor is limited by the bearing and the force fluctuation seen by the brush remains correspondingly small; the fraction of unbalance that reaches the electrical performance of the device is therefore markedly smaller than its mechanical effects.

3.3 Determination of Tolerances

3.3.1 Permissible residual unbalance

The objective is not to eliminate unbalance but to reduce it below a limit at which the functional requirements of the system are not threatened. This limit is termed the permissible residual unbalance. The standard recognises more than one route to establishing it: through balance quality grades, through a limited bearing force criterion, through experimental evaluation, and through established application experience [9]. The quality grade is only one of these routes and constitutes the weakest justification on its own where the component is not covered by the classification table.

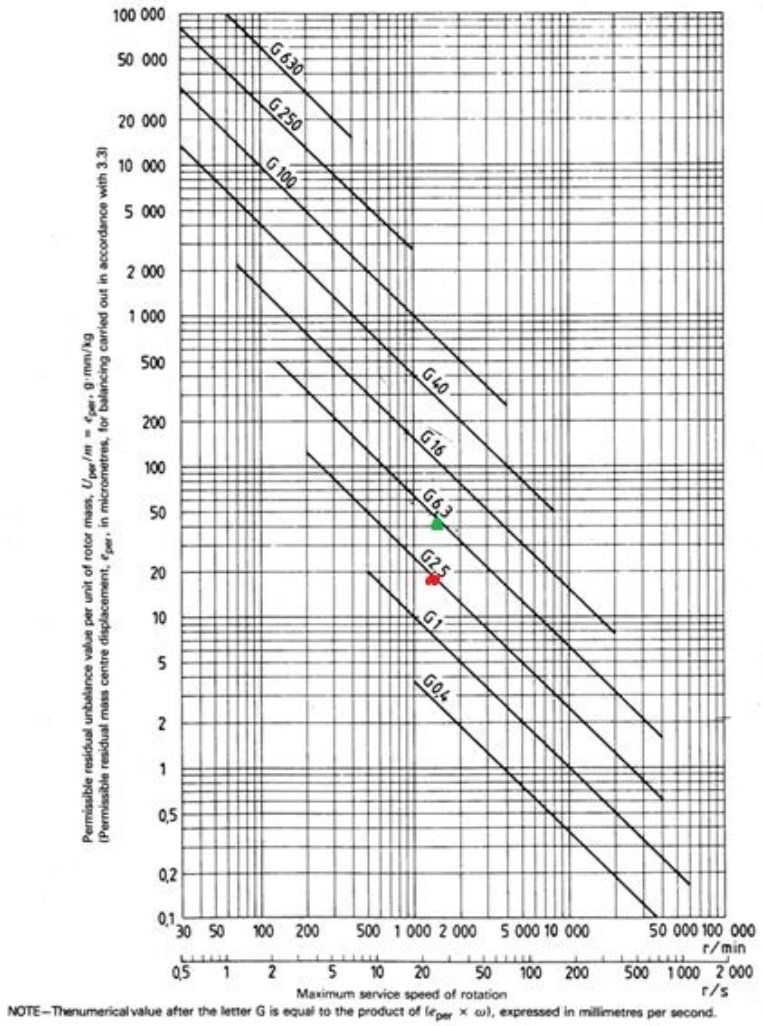
3.3.2 Balance quality grades

The standard classifies rotors into G grades according to their operating speed and the sensitivity of their application [9]. The G value is not a quality label but a physical quantity: the permissible orbital

velocity of the centre of mass, expressed in mm/s, $G = e_{per}\,\omega$. Defining the quantity as a velocity allows machines operating at different speeds to be assigned to the same grade, since where the orbital velocity of the centre of mass is equal, the dynamic behaviour is comparable.

Grade	Typical rotor types
G 0.4	Precision gyroscopes, precision grinding spindles, aerospace navigation systems
G 1.0	Magnetic disc drives, audio recording mechanisms, small armatures with special requirements
G 2.5	Gas and steam turbines, jet engine turbines, machine tool drives, armatures with special requirements
G 6.3	Standard electrical machine armatures, pump impellers, fan impellers, general machinery
G 16	Agricultural machinery, cardan shafts, crusher components
G 40	Automotive wheels and rims

Balance quality grades and the rotor categories to which they are conventionally assigned.



NOTE—The numerical value after the letter G is equal to the product of $(e_{per} \times \omega)$, expressed in millimetres per second.

3.3.3 The two-step calculation

Once the grade has been selected, two calculation steps follow, and these two are frequently confused. The first yields the permissible specific unbalance, a length expressing the permissible displacement of the centre of mass from the axis of rotation:

$$e_{per} = 1000\,G / \omega = 9549\,G / n_{max}$$

The constant 9549 arises from the conversion ω = 2πn/60. The second step yields the permissible residual unbalance, with the dimension g·mm:

$$U_{per} = e_{per}\, m$$

The speed in the first relation is always the maximum service speed and not the balancing speed; overlooking this distinction results in a tolerance calculated less restrictively than it should be. It should also be noted that G is not the vibration velocity measured on the housing, although both are expressed in mm/s; the two must be specified separately [28].

3.4 Translation into Manufacturing Tolerances

The permissible residual unbalance is a single number; in order to be transferred to manufacture it must be converted into geometric tolerances that can be expressed on a technical drawing.

3.4.1 The conversion chain and the mass ratio

The total runout measured with a dial indicator is twice the eccentricity, since the section centre traces a circle as the component is rotated and the indicator registers the diameter of that circle. The eccentricity is then multiplied by the mass that the error in question actually displaces:

$$\varepsilon = TIR / 2\,, \quad U_i = \varepsilon\, k_i\, m$$

The coefficient k_i expresses the fraction of the rotor mass displaced by an error on the surface concerned, and constitutes the entire logic of tolerance apportionment: a bearing seat carries a mass ratio of unity because an error there displaces the whole rotor, the body outer surface approximately 0.80, and an individual track approximately 0.05. Tolerances must therefore be graduated rather than uniform, with the tightest value assigned to the bearing seats and the loosest to the individual tracks.

3.4.2 Selection of the geometric symbol

The appropriate geometric tolerance for rotors is total runout, for three reasons. It measures directly the quantity that the balancing machine actually sees; it controls coaxiality and form errors together within a single tolerance callout; and it can be measured in the workshop using a dial indicator and vee blocks [29]. The concentricity symbol should be avoided, since its measurement requires a coordinate measuring machine and it has been largely abandoned in modern practice.

3.4.3 The datum axis

The axis about which the rotor turns in service is the axis defined by the bearing seats, and all geometric tolerances must be referenced to this axis. Selecting a different datum for manufacturing convenience introduces a discrepancy between the service axis and the inspection axis and invalidates the tolerance chain [30].

3.4.4 Machining in a single setup

A rule more effective than any tolerance value is that all surfaces required to be concentric must be machined in a single setup. In a repositioned setup the repeatability of the chuck or collet becomes the dominant error source irrespective of the accuracy of the machine tool; for a standard chuck this corresponds to a total runout of between fifty and one hundred fifty micrometres. The cost of applying the rule is zero and consists of a note on the technical drawing.

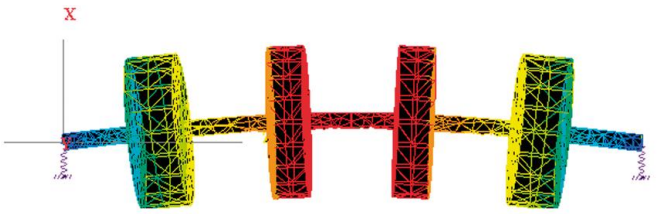
3.5 Rigid and Flexible Rotor Behaviour

A rotor is classified as having rigid behaviour if the deflection caused by unbalance is negligible in comparison with the agreed tolerance [10]. The notable aspect of this definition is that rigidity is not an absolute property of the rotor but a relation between the rotor and the tolerance: the same rotor may be classified as rigid under a loose tolerance and as flexible under a sufficiently restrictive one.

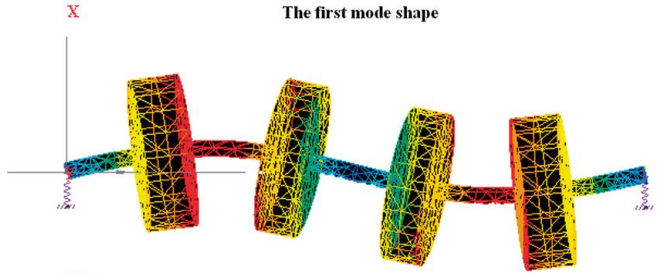
The practical equivalent is a question of feedback. If the rotor deflects by δ, part of its mass is displaced outward by that amount, which constitutes an additional eccentricity and therefore an additional unbalance, ΔU = m δ, so that ΔU / U = δ / e. The additional unbalance produces an additional force, which in turn produces further deflection; the rigid rotor assumption is the statement that this feedback is negligible.

If the rotor does not deflect, the unbalance state reduces to four quantities irrespective of how many heavy spots are present within it: amplitude and phase in each of two planes. Four unknowns and two correction planes yield an exact solution, and once the rotor has been balanced the result remains valid throughout the speed range. Where the rotor deflects, the axial location of the heavy spot becomes significant and the information no longer fits into four quantities; modal balancing is then applied, in which the rotor is balanced mode by mode and the number of correction planes required is the number of modes excited plus two [31].

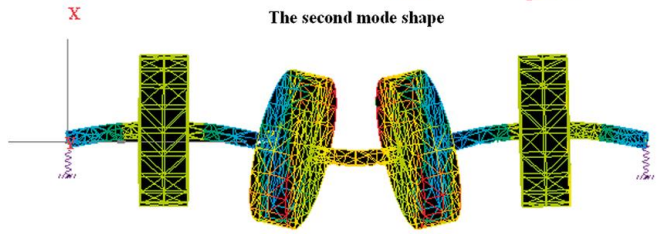
In short and stocky geometries the first critical speed corresponds not to the bending mode of the rotor but to the rigid-body mode of the rotor on its bearing support. Rigid rotor behaviour and rigid support are distinct concepts, and in such geometries the latter is generally the governing one.



The first mode shape



The second mode shape



The third mode shape

3.6 The Plane-Count Decision

3.6.1 The rule of thumb

A preliminary decision tool in widespread industrial use is based on the length-to-diameter ratio of the rotor: if the ratio is below one half the rotor is classified as disc-like and correction in a single plane may be sufficient, whereas above this value two correction planes are required. Three characteristics of the rule should be noted. No such threshold is defined in the standard, which bases the decision on the magnitude of the couple component relative to the tolerance [9]. The rule originates in industrial practice. And the threshold value varies between sources.

3.6.2 The limitation of the rule

The length-to-diameter ratio is a proxy variable representing the moment arm. It has two weaknesses: it gives the length of the arm but not whether mass is present on that arm, and it does not represent the inertial character of the rotor. The underlying assumption is that the component is a solid cylinder.

Couple unbalance, however, is a product-of-inertia quantity. The product of inertia arising when the rotor is mounted at an angle θ to the axis of rotation follows from the transformation of the inertia tensor:

$$I_{xz} = (I_d - I_p)\sin\theta\cos\theta \approx (I_d - I_p)\,\theta \quad (\theta \ll 1)$$

where I_p is the polar and I_d the transverse mass moment of inertia. For a hollow cylinder:

$$I_p = \tfrac{1}{2}\,m\,(R_o^2 + R_i^2), \quad I_d = \tfrac{1}{4}\,m\,(R_o^2 + R_i^2) + \tfrac{1}{12}\,m\,L^2$$

Three conclusions follow directly. If the bracket is zero the moment is zero, since a body for which $I_p = I_d$ is inertially spherical in that plane and every axis is a principal axis, so that no couple unbalance arises regardless of the misalignment angle. The sign of the bracket determines the behaviour: where I_p exceeds I_d the rotor is disc-like and the resulting moment acts so as to reduce the misalignment, whereas the reverse case is shaft-like and the moment increases it. And the length-to-diameter ratio

enters the relation through a single term, the $^1/_{12}$ m L² term within I_d, so that the rule observes only one component of the difference and takes no account of the wall ratio or of the radial distribution of mass. This is the reason that threshold values reported in the literature vary between sources.

3.6.3 The neutral ratio

The transition point at which the difference changes sign is found by equating the two moments of inertia. This operation does not express an assumption that the component is neutral, but determines the location of the transition boundary.

$$\tfrac{1}{2}\, m\,(R_o^{\,2} + R_i^{\,2}) = \tfrac{1}{4}\, m\,(R_o^{\,2} + R_i^{\,2}) + {}^1\!/_{12}\, m\, L^2$$

The mass cancels, so that the boundary is a purely geometric quantity:

$$\tfrac{1}{4}\,(R_o^{\,2} + R_i^{\,2}) = {}^1\!/_{12}\, L^2 \;\Longrightarrow\; L^2 = 3\,(R_o^{\,2} + R_i^{\,2})$$

Dividing by D = 2R_o yields a closed form:

$$(L/D)_{neutral} = (\sqrt{3}/2)\,\sqrt{(1 + (R_i/R_o)^2)}$$

The result demonstrates that the neutral ratio is not a fixed value but varies with the wall ratio between 0.866 for a solid cylinder and 1.225 for a thin-walled tube in the limiting case. The fixed thresholds employed in the literature are approximations selected on the conservative side of this variable boundary.

This criterion evaluates sensitivity to angular misalignment only. The remaining sources of couple unbalance — the relative misalignment of the bearing seats and axial mass asymmetry — must be calculated separately, so that the neutral ratio comparison does not by itself determine the plane count.

3.6.4 The verification criterion

The decision must be tied to a measurable criterion. Following balancing, amplitude and phase are read at both bearings and the components are decomposed:

$$U_s = U_L + U_R\,, \quad U_c = (b/2)\,|\,U_L - U_R\,|$$

A phase difference near zero indicates that the static component is dominant, and a difference near one hundred eighty degrees that the couple component is dominant. If the decomposed couple component exceeds the plane tolerance, two correction planes are adopted. This measurement constitutes the documentary record of the decision [32].

3.7 Measurement and Correction Methods

3.7.1 Quantities to be obtained from measurement

Exactly four quantities must be obtained from the balancing machine: the amplitude and phase of the first-order vibration, measured at two separate locations. Without the phase angle the magnitude of the unbalance can be determined but not its angular position, so that the location at which the correction mass is to be applied cannot be established. With measurement at a single location a total value is obtained, but it cannot be determined whether the problem is static or couple in nature.

3.7.2 Machine architectures

Two types of balancing machine are used in industry, and the nomenclature is misleading: what is soft or hard is not the bearing on which the rotor rests but the suspension that carries that bearing [33].

In soft-bearing machines the rotor rests on a movable cradle supported on flexible springs, whose natural frequency lies appreciably below the balancing speed so that the system operates in the supercritical region. Under this condition the angular velocity terms cancel in the equation of motion:

$$X = m\,e\,\omega^2 / [\,M\,(\omega_n{}^2 - \omega^2)\,] \;\rightarrow\; -U/M \quad (\omega \gg \omega_n)$$

The displacement read by the machine therefore becomes independent of rotational speed, which is why such machines achieve high sensitivity even at low and therefore safe speeds. The cost is that the machine cannot determine the axial location of the correction planes, which requires a calibration run with a trial mass for every new rotor type.

In hard-bearing machines the suspension is rigid and the machine measures the centrifugal force directly by means of piezoelectric force transducers embedded within the supports. Because the system is predictable, no trial mass is required; the geometry is entered and the machine computes the coefficient itself, a property termed permanent calibration. In practice the most reliable means of distinguishing the two is to observe the workflow: if a trial mass is applied the machine is soft-bearing, and if geometry is entered it is hard-bearing.

Both types lose sensitivity at the lower end of their nominal mass range, though by different mechanisms. In hard-bearing machines the load cells are sized to the capacity of the machine, so small forces fall near the resolution limit; in soft-bearing machines the sensitivity is determined by the combined mass of the rotor and the cradle, so a heavy cradle attenuates the signal proportionally.

3.7.3 Measurement elements

Element	Measured quantity	Principle	Machine type
Seismic velocity transducer	Vibration velocity	Spring-mounted magnet within a fixed coil; induced voltage	Soft-bearing; mounted outside the support
Piezoelectric force transducer	Centrifugal force	Crystal in the load path; charge separation under compression	Hard-bearing; embedded within the support
Accelerometer	Vibration acceleration	Seismic mass compressing a piezoelectric crystal	Field balancing; mounted on the housing
Optical tachometer	Phase reference and speed	Reflective strip producing one pulse per revolution	Both types; mandatory

Measurement elements employed in balancing machines, with their operating principles and typical application.

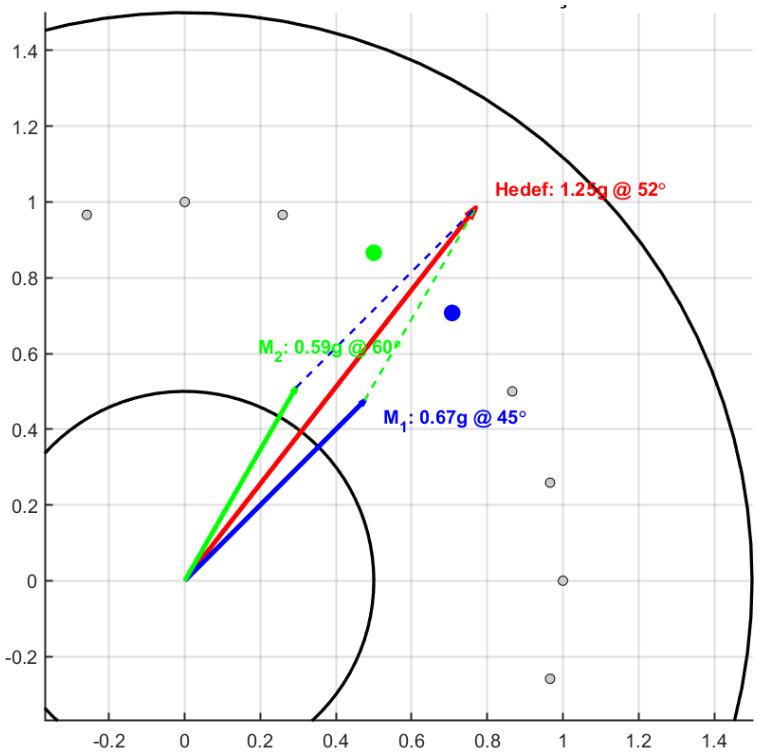
The velocity transducer and the accelerometer require motion in order to produce a signal, whereas the force transducer does not. The accelerometer and the force transducer are two applications of the same piezoelectric principle, distinguished by the presence of a known seismic mass on the crystal in the former. The optical tachometer answers a different question from the other three: the amplitude transducers establish how much unbalance is present, while the tachometer establishes where it is.

3.7.4 Correction methods

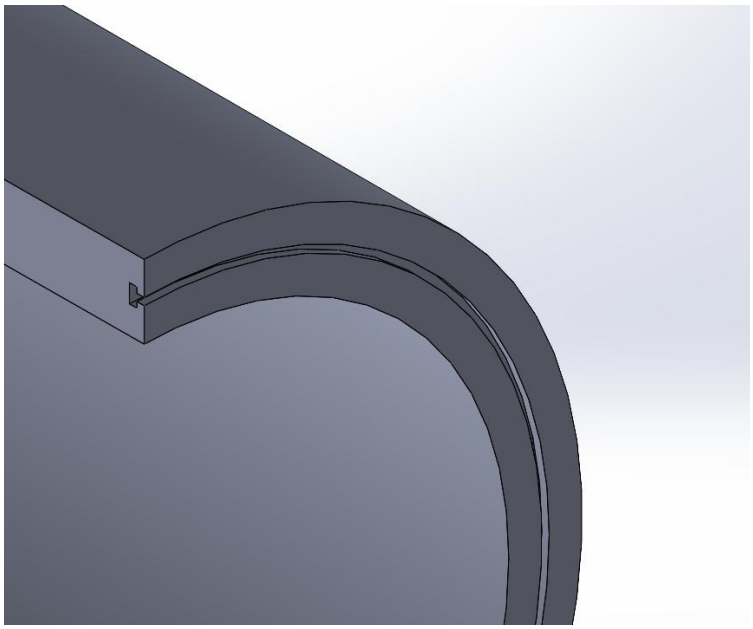
The measured unbalance may be corrected by removing mass, by adding mass, or by adjusting the position of a mass.

Material removal. Material is removed at the angular position of the heavy spot, generally by drilling, with $m = U / r_d$. No component is added to the system, so the configuration does not change over the service life and no retention requirement arises. The method is not reversible, however, which conflicts with the iterative nature of balancing, and in slip ring applications the conductive swarf constitutes an electrical hazard. The reduction in total mass also narrows the permissible unbalance, which is proportional to mass.

Threaded correction masses. Threaded holes are provided in the correction plane at a fixed radius and at equal angular intervals. Although the holes lie at fixed angles while the target angle generally falls between two of them, this limitation is eliminated by vector splitting: the target mass is divided between the two holes bracketing the target in proportions obtained from the sine rule, $m_1 = m_t \sin(\phi_2 - \phi) / \sin(\phi_2 - \phi_1)$, so that the resultant lies exactly at the target angle. The residual error therefore arises entirely from mass quantisation, $\Delta U = r_d \, \delta m$, which means that the benefit of increasing the number of holes diminishes rapidly and the effective route to better resolution is a more precise balance rather than more holes. The method offers the strongest traceability, since the correction can be recorded as a hole number and a mass, but the hole pattern must be fixed at the design stage while the unbalance distribution is still unknown.



T-slot. A continuous circular slot of T-shaped section is machined into the correction plane, and a sliding block within the slot is moved to the calculated position and secured by friction and adhesive. The principal advantage is not angular resolution, which vector splitting already provides, but design flexibility: the slot is continuous, the block mass is free and the position is free, so that uncertainty is transferred from the design stage to the operational stage. The undercut geometry requires a dedicated form tool and an additional milling operation, the block is retained by friction rather than thread engagement so that a separate retention provision is required, and traceability is poor.



Criterion	Material removal	Threaded mass	T-slot
Reversibility	None	Partial	Full
Angular resolution	Continuous	Exact by vector splitting	Continuous
Mass resolution	Continuous	Discrete	Continuous
Manufacture	Standard	Standard	Form tool, extra operation
Retention	Not required	Thread engagement	Separately specified
Traceability	Poor	High	Poor
Design flexibility	High	Low	High
Application-specific risk	Conductive swarf	Stress concentration	Block displacement

Comparison of the three correction methods against the criteria that govern the selection.

3.8 Component Description and Requirements

The component examined is the rotor of a slip ring assembly of tubular cross-section.

Parameter	Symbol	Value
Mass	m	0.700 kg
Inner radius	R_i	50 mm
Outer radius	R_o	60 mm
Length	L	70 mm
Material	—	Aluminium, E = 70 GPa
Operating speed	n	1500 rev/min
Correction radius	r_d	55 mm

Principal parameters of the rotor examined in this study.

$$\omega = 2\pi n / 60 = 157.08\ \text{rad/s}, \quad L/D = 70/120 = 0.583$$

The rotor carries the conducting tracks on its outer surface and the internal wiring within its bore, and is supported on two rolling-element bearings. The assembly operates continuously at its nominal speed, so the analysis addresses a single operating point throughout.

3.9 Balance Quality Grade Selection

As established in Section 2.5.4, no balance quality grade is defined for slip rings. The grade was therefore established by analogy, and the closest category in the classification is the armature of an electrical machine. The correspondence rests on four points: a rotating body carrying concentric conducting surfaces, stationary brushes sliding on those surfaces, a comparable speed band, and an identical governing failure mechanism in contact stability.

Two differences were identified, and both fall on the conservative side. An armature operates across a magnetic air gap of a few hundred micrometres, so that unbalance modulates that gap and produces a one-sided magnetic pull which reinforces the initial displacement; no magnetic circuit exists in the present component. An armature commutator is segmented, so the brush traverses insulating gaps at every revolution and is subject to mechanical impact and commutation arcing independently of unbalance; the tracks of the present component are continuous. The analogue therefore operates under a more severe loading than the component to which the grade is transferred.

The grade for standard armatures is G 6.3, from which:

$$e_{per} = 9549\,G / n = (\,9549 \times 6.3\,) / 1500 = 40.1\ \mu m$$

$$U_{per} = e_{per}\,m = 40.1\ \mu m \times 0.700\ kg = 28.1\ g{\cdot}mm$$

$$m_{corr} = U_{per} / r_d = 28.1 / 55 = 0.51\ g$$

A single screw of half a gram therefore consumes the entire tolerance.

The adjacent grades were eliminated. G 16 provides no discriminating acceptance criterion, since the corresponding tolerance of 71.3 g·mm lies within the range of the assembly contributions quantified in Section 3.12, so a limit at that level would reject nothing. G 2.5 changes the manufacturing process rather than the balancing operation, requiring a transition from standard turning to precision turning or grinding in exchange for an improvement that cannot be measured at the brush contact. The grade is nevertheless adopted as a starting value rather than as a conclusion.

3.10 Verification of the Rigid Rotor Assumption

Following the definition in ISO 21940-1 [10], the governing question is whether the rotor deflects sufficiently in rotation to augment its own unbalance. The unbalance force at the tolerance limit is

$$F = U_{per}\,\omega^2 = 2.81 \times 10^{-5}\,kg{\cdot}m \times (\,157.08\,)^2 = 0.69\ N$$

The second moment of area of the tubular section is

$$I = (\,\pi / 4\,)\,(\,R_o^{\,4} - R_i^{\,4}\,) = 5.27 \times 10^{6}\,mm^4$$

Treating the rotor as a simply supported beam loaded at mid-span,

$$\delta = F\,L^3 / (\,48\,E\,I\,) = 1.3 \times 10^{-11}\,m$$

and the additional unbalance produced by this deflection, relative to the existing unbalance, is

$$\Delta U / U = \delta / e_{per} = 1.3 \times 10^{-11} / 40.1 \times 10^{-6} = 3.3 \times 10^{-7}$$

The deflection augments the unbalance by three parts in ten million; no feedback exists and the rotor behaves rigidly.

The Euler–Bernoulli beam model is derived for slender beams and neglects shear deformation and rotary inertia, so that at L / D = 0.583 these assumptions are weak and the model understates the deflection. The margin is of such magnitude, however, that a correction of several times does not alter the conclusion, and the argument rests on the deflection ratio rather than on an absolute value.

In this geometry the first critical speed corresponds to the rigid-body mode of the rotor on its bearing support rather than to the bending mode. Taking a conservative effective radial stiffness, the conical mode is of the order of 57,500 rev/min, some thirty-eight times the operating speed, and the dynamic amplification factor is 1 / (1 − r²) = 1.0007 for r = 0.026. Reducing the assumed support stiffness by two orders of magnitude leaves the natural frequency at approximately four times the operating speed, so the assumption is insensitive to an error of two orders of magnitude in its inputs.

3.11 Plane-Count Decision and the Inertia Criterion

The length-to-diameter ratio of the component is 0.583, marginally above the conventional threshold of 0.5, so that the rule of thumb indicates two correction planes. The mass moments of inertia were evaluated:

$$I_p = \tfrac{1}{2}\, m\,(R_o^{\,2} + R_i^{\,2}) = 2.14 \times 10^{-3}\ \mathrm{kg\cdot m^2}$$

$$I_d = \tfrac{1}{4}\, m\,(R_o^{\,2} + R_i^{\,2}) + \tfrac{1}{12}\, m\, L^2 = 1.35 \times 10^{-3}\ \mathrm{kg\cdot m^2} \;\Longrightarrow\; I_p / I_d = 1.58$$

The ratio exceeds unity, so the rotor is disc-like in character and the moment arising from angular misalignment acts so as to reduce rather than to increase that misalignment. The neutral ratio for this cross-section, from the closed form derived in Section 3.6.3, is 1.13, against an actual ratio of 0.583 — approximately half of the transition value. Sensitivity to angular misalignment is therefore limited. This comparison establishes only that the component lies on the insensitive side; it does not by itself determine the plane count.

Three sources of couple unbalance were evaluated. With a relative misalignment of the bearing seats of 12.5 µm over a bearing span of 40 mm, the tilt angle is θ = 3.13 × 10⁻⁴ rad, giving $U_{c,1} = |\, I_d - I_p \,|\, \theta$ = 244 g·mm². A static contribution of 32 g·mm from the internal wiring, located 25 mm from the plane of the centre of mass, gives $U_{c,2}$ = 800 g·mm². Combining:

$$U_c = \sqrt{(\,244^2 + 800^2\,)} = 837\ \mathrm{g\cdot mm^2} \;\Longrightarrow\; U_c / b = 837 / 70 = 11.96\ \mathrm{g\cdot mm}$$

against a plane tolerance of 14.04 g·mm, that is 85 percent of the permissible value, with the wiring accounting for 96 percent of the squared contribution.

Single-plane balancing was adopted. The margin is narrow, so the decision is presented as conditional rather than as an unqualified statement of sufficiency, and two conditions are attached: the internal wiring and connectors must be distributed symmetrically in the axial direction, and the correction must be applied in the plane of the centre of mass, since a correct mass placed at an incorrect axial location generates more couple unbalance than it removes. The decision is further tied to the verification criterion of Section 3.6.4, and no separate measurement run is required since a two-plane balancing machine displays the amplitude and phase of each plane directly.

3.12 Manufacturing and Assembly Error Budget

This section constitutes the principal result of the study. Its purpose is to establish whether the tolerance derived in Section 3.9 is attainable in production, and the answer proves to depend on contributions that are not controlled by the technical drawing.

Each error source was converted into an unbalance contribution through the chain established in Section 3.4.1. Since the contributions are vector quantities whose angular orientations are mutually independent, they are combined as a root-sum-square rather than arithmetically. The arithmetic sum represents the worst case, in which every contribution aligns at the same angle, and is retained as an upper bound but not used as the design value.

#	Source	Tolerance	k_i	ε	U_i [g·mm]	Share
1	Bearing seat concentricity	0.03 mm	1.00	15 µm	10.50	44.1 %
2	Body runout to bearing axis	0.03 mm	0.80	15 µm	8.40	28.2 %
3	Assembly repeatability	0.02 mm	1.00	10 µm	7.00	19.6 %
4	Bearing inner ring runout (P0)	0.01 mm	1.00	5 µm	3.50	4.9 %
5	Individual tracks (8 off, RSS)	0.05 mm	0.05 ea.	25 µm	2.48	2.5 %
6	Material density inhomogeneity	—	1.00	2 µm	1.40	0.8 %

Unbalance contributions of the manufacturing and assembly error sources, combined as a root-sum-square.

$$U_{manufacturing} = \sqrt{(\,10.50^2 + 8.40^2 + 7.00^2 + 3.50^2 + 2.48^2 + 1.40^2\,)} = 15.8\ g{\cdot}mm$$

This is 56 percent of the permissible residual unbalance of 28.1 g·mm; considered in isolation, manufacturing is comfortably within tolerance. The first three items account for 92 percent of the squared total, which confirms the graduated tolerance logic of Section 3.4.1: the bearing seats, carrying a mass ratio of unity, dominate, while the individual tracks contribute 2.5 percent despite carrying the loosest tolerance of the set.

Two items deserve separate comment. Assembly repeatability is not a drawing tolerance but a process characteristic, determined by fixture design, tightening torque and the discipline of referencing; it cannot be improved by specifying a tighter value on a drawing, and it recurs in Section 3.14 as the dominant term of the measurement uncertainty budget. Material density inhomogeneity is independent of machining accuracy and is unaffected by tightening any tolerance, for which reason it is held constant in the comparison below.

Three further contributions were identified which cannot be expressed as a geometric tolerance.

#	Source	U [g·mm]	Basis of estimate
7	Wiring asymmetry	10 – 40	0.3 g conductor at 55 mm ≈ 16 g·mm per conductor

#	Source	U [g·mm]	Basis of estimate
8	Re-fixturing during machining	10 – 50	Chuck repeatability 50–150 µm TIR
9	Bonding and fixing elements	20 – 50	0.5 g adhesive at 55 mm ≈ 27 g·mm

Assembly contributions that cannot be expressed as drawing tolerances, with the basis of each estimate.

These three items differ from the preceding six in kind rather than in magnitude. A total runout of 0.03 mm can be specified on a drawing and verified by inspection; the side of the rotor onto which the internal wiring is routed, the quantity of adhesive that flows to one side during curing, and whether a second machining setup is required cannot be specified in the same way. They are governed by the assembly instruction and by the process plan. It should be emphasised that these are engineering estimates rather than measured values, a limitation declared in Section 1.4.

Combining all nine contributions, using mid-range values for items 7 to 9:

$$U_{total} = \sqrt{(250 + 25^2 + 30^2 + 35^2)} = \sqrt{3000} = 54.8\ g{\cdot}mm$$

This is 1.95 times the permissible residual unbalance. Taking the optimistic bounds of the three estimates gives 29.2 g·mm, marginally above tolerance; taking the pessimistic bounds gives 82.8 g·mm. Under no combination of plausible values does the assembled unit fall within tolerance without correction, so that balancing is a production operation rather than an optional quality improvement.

A second scenario was constructed in which all machining tolerances of items 1 to 5 were reduced by approximately one half, representing a transition to precision machining, with item 6 held constant. The manufacturing contribution falls from 15.8 to 8.0 g·mm, an improvement of 49 percent at that stage. Adding the same assembly contributions, however:

$$U_{total,\ tight} = \sqrt{(64 + 625 + 900 + 1225)} = 53.0\ g{\cdot}mm \quad against \quad 54.8\ g{\cdot}mm$$

The entire benefit of precision machining on the final result is 3.3 percent. This follows from the nature of root-sum-square combination: reducing a term that is already small is concealed by the larger terms beside it. Tightening the drawing tolerances of this component would therefore incur the cost of a changed manufacturing process in exchange for an improvement that could not be resolved by measurement.

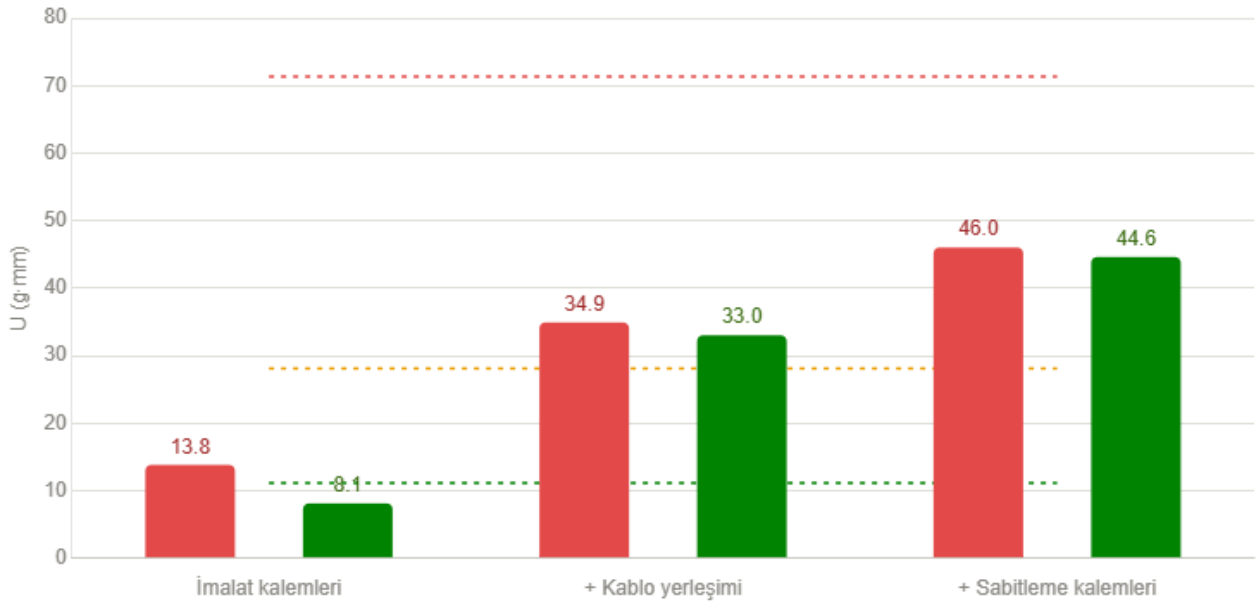
Since the dominant contributions are governed by the assembly process, three interventions were identified which act on them directly. The net contribution of the internal conductors is proportional to the difference between the number routed to one side and the number routed to the opposite side, $U_{net} = | N_{0°} - N_{180°} |\ u$; eleven conductors routed entirely to one side contribute approximately 45 g·mm, whereas split six and five the difference reduces to one and the net contribution falls to approximately 4 g·mm, a reduction of 91 percent. Machining all concentric surfaces in a single setup eliminates item 8 entirely. Symmetric placement of the bonding and fixing elements reduces item 9 substantially, since the residual contribution is the asymmetry rather than the mass itself. Applying all three, and retaining the original manufacturing tolerances:

$$U_{total,\ improved} = \sqrt{(250 + 16 + 0 + 144)} = 20.3\ g{\cdot}mm$$

This lies within the permissible residual unbalance of 28.1 g·mm, at 72 percent of the limit, and is achieved at essentially zero cost, since each measure consists of a line in an assembly instruction or a

note on a drawing. The comparison with the 3.3 percent obtained by precision machining is the central result of this study.

Should measured values persist outside tolerance in production, the angular positions of the applied corrections are to be recorded for the first five to ten units. Randomly distributed angles indicate manufacturing variation, so that every unit requires balancing; angles clustering within a defined sector indicate a systematic design or assembly feature, which is corrected once in design and thereafter for the entire production run. This distinction cannot be drawn from the magnitude of the unbalance alone.



3.13 Balancing Machine Selection

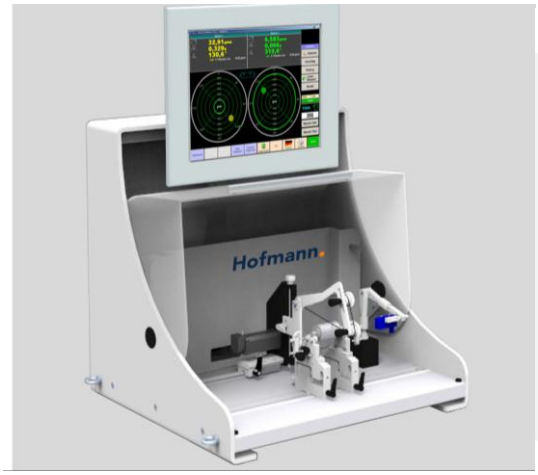
Three machines were available: two of hard-bearing and one of soft-bearing architecture. An adequacy criterion was established requiring the minimum achievable residual unbalance of the machine to lie below one quarter of the plane tolerance, that is below 3.51 g·mm.

Machine	Architecture	e_{mar} [g·mm/kg]	U_{mar} [g·mm]	Margin
Schenck Pasio 5	Hard-bearing	0.15	0.105	33 ×
Hofmann H-9.1	Hard-bearing	0.10	0.070	50 ×
IRD B01/S	Soft-bearing	0.015	0.011	334 ×

Minimum achievable residual unbalance of the candidate machines, evaluated against the adequacy criterion.

All three satisfy the criterion with substantial margin, so machine sensitivity is not a governing constraint in this application. The selection therefore rests on two further considerations. The first is the mass and diameter range: at 0.700 kg the component lies in the lower portion of the nominal mass range of all three machines, where both architectures lose sensitivity, and the outer diameter of 120 mm must lie within the maximum rotor diameter of the machine. The second, and the determining criterion, is fixturing: the machines support the rotor on its own journals and their journal diameter ranges differ, and one machine requires an auxiliary shaft to reach the journal diameter of the present component, which introduces an additional uncertainty term and an additional concentricity error into the measurement chain. That machine was eliminated.

A requirement independent of machine architecture is that the measurement must be performed on the fully assembled unit. The wiring and the bonding elements are added after machining and constitute the dominant contributions of the error budget, so that balancing the bare rotor would exclude precisely those contributions from the measurement.





3.14 Measurement Uncertainty and Acceptance

A measured value is not accepted directly, since the measurement itself carries an uncertainty. Four contributions were identified and combined as a root-sum-square: the resolution of the machine, the repeatability of the measurement, thermal drift, and the placement of the correction mass [39].

Repeatability is quantified by mounting, measuring, dismounting and remounting the same unit ten times without alteration, and taking the standard deviation of the results. Approximately three quarters of the total uncertainty arises from this term, while the contribution of the machine resolution is negligible. The practical consequence is that the route to reducing measurement uncertainty is fixturing discipline rather than the procurement of a more sensitive machine — the same conclusion, arrived at independently, as that of Section 3.12 regarding manufacturing tolerances.

Three acceptance levels were defined, with the uncertainty allowance subtracted from within the tolerance rather than added to it, so that a less certain measurement results in a tighter acceptance limit:

$$U_{target} < U_{accept} = U_{per} - u_{total} < U_{spec}$$

Three independent verification procedures were specified. In the angular scan a known mass is fitted at eight angular positions in turn at the correction radius; the machine should report the same magnitude at every position and an angle matching the fitted position, which verifies angular accuracy and linearity. In the speed scan the unit is measured at 500, 1000 and 1500 rev/min; the reported value in g·mm should remain constant, since unbalance is independent of speed, and a constant result is the empirical verification of the rigid rotor assumption of Section 3.10. In the thirty-minute run the unit is run at service speed and re-measured, which captures settling of the fasteners and bonding elements, thermal drift, and retention of the correction block.

3.15 Correction Method Selection

The three methods of Section 3.7.4 yield comparable resolution for this component, so the selection was made on the remaining criteria. Material removal was eliminated on two grounds: it is irreversible,

which conflicts with the iterative nature of balancing, and aluminium swarf entering the internal volume would reduce the insulation resistance between channels.

Of the two remaining methods, the comparison favours threaded correction masses on the criteria of manufacturing cost, retention and traceability. This is acknowledged rather than contested. The T-slot was nevertheless selected, on the ground of design flexibility. The component is at prototype stage, and the wiring layout, the volume of the bonding elements and the required correction capacity are not yet fixed; a threaded hole pattern requires the hole count, thread size, depth, radius and prohibited sectors to be determined while the unbalance distribution remains unknown, and any subsequent change requires the component to be re-drilled. The T-slot transfers this uncertainty from the design stage to the operational stage.

The cost of the selection is accepted and recorded in the specification: a dedicated form tool for the undercut, an additional milling operation, and a separately specified retention provision for the block. The slot is to be machined into the axial end face rather than the circumferential surface, since the circumferential surface carries the conducting tracks and the sealing surfaces and offers no free area, and a circumferential slot would confine correction to a single axial plane. Once the internal layout is fixed and the component enters series production, a reversion to threaded correction masses may be the appropriate decision.

3.16 Findings and Open Items

Four results are reported from this project.

The conventional length-to-diameter threshold is not a fixed value. A closed-form expression for the neutral ratio was derived from the difference between the polar and transverse mass moments of inertia, showing that the transition between disc-like and shaft-like behaviour occurs between 0.866 and 1.225 depending on the wall ratio. For the present component the transition lies at 1.13 against an actual ratio of 0.583. The plane-count decision was accordingly based on the couple unbalance budget rather than on the rule of thumb, and was issued conditionally.

Manufacturing precision is not the governing contribution. The machining and assembly tolerances together produce 15.8 g·mm against a permissible value of 28.1 g·mm, while the contributions that cannot be expressed as drawing tolerances raise the total to approximately 54.8 g·mm. Halving every machining tolerance improves the final result by 3.3 percent.

Three interventions at zero cost bring the assembly within tolerance. Distributing the internal conductors symmetrically, machining all concentric surfaces in a single setup, and placing the bonding elements symmetrically together reduce the total to 20.3 g·mm.

The dominant term of the measurement uncertainty is fixturing, not instrumentation. All three candidate machines exceed the adequacy criterion by a factor of at least thirty, while approximately three quarters of the measurement uncertainty arises from the repeatability with which the unit seats on the machine.

A common pattern connects the second and fourth of these results, and it is the principal observation of the study: in each case the quantity that receives the most engineering attention is not the quantity that governs the outcome.

Three questions remain unresolved. The values assigned to the wiring, re-fixturing and bonding contributions are engineering estimates rather than measurements, and these are the dominant terms of the budget; a measurement on the first assembled unit is required to replace them. No balancing measurement was performed on the physical component during the internship period, so all results are analytical and the phase decomposition on which the plane-count decision depends has not been carried out. And the distribution of the correction angles across units is unknown, which determines whether every unit requires balancing or whether a single design correction resolves the matter for the production run.

4. Project 2 — Design and Manufacture of an Ergonomic Tool Holder

4.1 Problem definition

A second design task was assigned during the internship, independent of the balancing study. At the assembly and soldering station in the laboratory, hand tools were kept loose on the work surface. Screwdrivers, soldering syringes and similar implements accumulated without a defined position, which produced two consequences: the working area available for the assembly itself was progressively reduced, and the time spent locating a required tool between operations was greater than the operation warranted.

The task was to design a holder that would give each tool a defined position without consuming the working surface.

4.2 Design requirements

Ergonomic orientation. The tools are retrieved and returned repeatedly during an assembly sequence, so the holder must present them at an orientation consistent with the natural motion of the operator's hand rather than requiring a deliberate reaching movement. An inclined arrangement was adopted, so that the handles of the tools are presented towards the operator and may be grasped without the wrist departing from its neutral position.

Preservation of the working area. A holder that occupies the surface it is intended to clear defeats its own purpose. The holder was therefore designed for concealed mounting by a single screw from beneath the work surface, which leaves no fastener, bracket or footprint on the upper surface and permits the holder to be positioned at the edge of the working area rather than within it.

4.3 Design and manufacture

The holder was modelled in Siemens NX. The geometry consists of an inclined body carrying individual pockets sized to the tools in use, a base plate incorporating the single mounting feature, and a fillet transition between the two to avoid a stress concentration at the junction, since the holder is loaded as a cantilever when a tool is withdrawn.

The component was produced by additive manufacturing as a single piece. This process was appropriate to the task for three reasons: the quantity required was one, the geometry contains

inclined pockets that would require multiple setups in a subtractive process, and the pocket dimensions could be adjusted between iterations at negligible cost.

4.4 Outcome

The holder was manufactured and placed in service at the station for which it was designed.

[Buraya kendi değerlendirmeni ekle: kullanıma alındı mı, geri bildirim geldi mi, ikinci bir iterasyon gerekti mi.]

The task differed from the balancing study in character rather than in discipline. The balancing work proceeded from a specification towards a quantitative tolerance; the holder proceeded from an observed inconvenience towards a physical object, with the requirements established by the designer rather than supplied. Both, however, resolved into the same question addressed throughout this report: identifying which constraint actually governs the outcome, and directing the design effort towards that constraint rather than towards the one that is easiest to quantify.

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